

Contoso Ltd. -- VPN Configuration Manual

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1. Overview

This manual provides instructions for configuring and using the Contoso corporate VPN. The VPN is required for accessing internal resources (intranet, file shares, internal applications) when working outside the corporate office network.

Contoso uses Palo Alto GlobalProtect as the VPN client.

2. System Requirements

Requirement	Specification
Operating System	Windows 10/11 Enterprise, macOS 13+
VPN Client	GlobalProtect 6.1.x or later
Network	Broadband internet (minimum 10 Mbps reco
Authentication	Corporate credentials + MFA (Microsoft A

3. Installation

3.1 Windows

GlobalProtect is pre-installed on Contoso-managed devices via Intune. If the client is missing:

- Open Company Portal.
- Search for "GlobalProtect."
- Click Install.
- Wait for installation to complete (approximately 3-5 minutes).
- Restart your laptop if prompted.

3.2 macOS

- Open Company Portal on your Mac.
- Search for "GlobalProtect."
- Click Install and approve the system extension when prompted.
- Go to System Settings > Privacy & Security > Network Extensions and allow "Palo Alto Networks."
- Restart your Mac.

4. Connection Setup

4.1 First-Time Configuration

Launch GlobalProtect from the system tray (Windows) or menu bar (macOS).

Enter the portal address: vpn.contoso.com

Click Connect.

Enter your corporate email and password.

Approve the MFA prompt on your Microsoft Authenticator app.

The client will download the VPN configuration automatically.

4.2 Connection Modes

Contoso VPN supports the following modes:

- Full Tunnel (Default): All traffic is routed through the VPN. This is the default mode for all employees.
- Split Tunnel: Only traffic destined for Contoso internal networks (10.0.0.0/8, 172.16.0.0/12) is routed through the VPN. Personal browsing goes directly to the internet.

Note: Split tunnel mode is available for all employees in v1 2025. To enable split tunnel:

Open GlobalProtect settings.

Go to Connection > Tunnel Mode.

Select Split Tunnel.

Click Apply.

4.3 Always-On VPN

For users in the Finance and Legal departments, Always-On VPN is enforced by policy. The VPN connects automatically when an internet connection is detected and cannot be manually disconnected.

5. Authentication

5.1 Credentials

- Username: Your corporate email address (e.g., john.doe@contoso.com)
- Password: Your corporate Active Directory password

5.2 Multi-Factor Authentication

MFA is required for all VPN connections. Supported MFA methods:

Microsoft Authenticator push notification (recommended)

TOTP code from Microsoft Authenticator

SMS code (backup only -- must be pre-registered)

5.3 Certificate-Based Authentication

Certificate-based authentication is not available in v1 2025. All users must authenticate with username/password + MFA.

6. Network Settings

6.1 MTU Configuration

The default MTU size is 1500. If you experience connectivity issues (frequent disconnections, slow performance), you may need to adjust:

- Recommended MTU for VPN: 1400
- To change MTU on Windows:

```
netsh interface ipv4 set subinterface "GlobalProtect" mtu=1400 store=persistent
```

- To change MTU on macOS:

```
sudo ifconfig utun0 mtu 1400
```

6.2 DNS Settings

When connected to the VPN:

- Primary DNS: 10.1.1.10
- Secondary DNS: 10.1.1.11
- DNS suffix: corp.contoso.com

These are configured automatically. Do not manually change DNS settings.

6.3 IP Address Assignment

VPN clients receive IP addresses from the 10.200.0.0/16 range via DHCP. IP addresses are dynamically assigned and may change between sessions.

7. Session Management

- Session timeout: 8 hours of continuous connection.
- Idle timeout: 30 minutes of no network activity.
- After timeout, you must reconnect and re-authenticate.
- Maximum concurrent sessions: 1 per user account.

8. Troubleshooting

8.1 Cannot Connect

Symptom	Resolution
"Portal unreachable" error	Check internet connection. Try accessing
"Authentication failed"	Verify your password hasn't expired. Res
MFA prompt not received	Open Microsoft Authenticator and check f
Connection rejected	Your device may not be compliant. Open C

8.2 Frequent Disconnections

If the VPN disconnects frequently:

Lower MTU to 1400 (see Section 6.1).

Check your internet stability -- run a speed test at <https://speedtest.net>.

If on Wi-Fi, try switching to a wired connection.

Disable any personal VPN or proxy software.

If the issue persists, collect GlobalProtect logs:

- Open GlobalProtect > Settings > Troubleshooting > Collect Logs.
- Submit the log file to the IT Helpdesk.

8.3 Slow Performance

- Verify your internet speed meets the 10 Mbps minimum requirement.
- If using full tunnel mode, consider requesting split tunnel access.
- Close bandwidth-intensive applications (video streaming, large file downloads).
- Contact network team if internal application response times exceed 500 ms.

9. Security Policies

- VPN access is restricted to Contoso-managed and compliant devices only.
- Jailbroken or rooted devices are automatically blocked.
- VPN connection logs are retained for 90 days for audit purposes.
- Sharing VPN credentials is a violation of IT security policy and grounds for disciplinary action.

10. Support

For VPN issues, contact the IT Helpdesk:

- IT Portal: <https://itportal.contoso.com>
- Email: helpdesk@contoso.com
- Phone: +1 (555) 100-2000 (Option 3 for network issues)
- Priority: VPN outages for remote workers are treated as Priority 2 (High).

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